

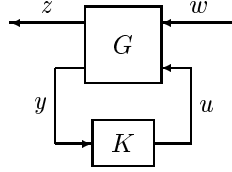
17

$\mathcal{H}_\infty$ Control: General Case

In this chapter we will consider again the standard $\mathcal{H}_\infty$ control problem but with some assumptions in the last chapter relaxed. Since the proof techniques in the last chapter can be applied to this general case except with some more involved algebra, the detailed proof for the general case will not be given; only the formulas are presented. However, some procedures to carry out the proof will be outlined together with some alternative approaches to solve the standard $\mathcal{H}_\infty$ problem and some interpretations of the solutions. We will also indicate how the assumptions in the general case can be relaxed further to accommodate other more complicated problems. More specifically, Section 17.1 presents the solutions to the general $\mathcal{H}_\infty$ problem. Section 17.2 discusses the techniques to transform a general problem to a standard problem which satisfies the assumptions in the last chapter. The problems associated with relaxing the assumptions for the general standard problems and techniques for dealing with them will be considered in Section 17.3. Section 17.4 considers the integral control in the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ theory and Section 17.5 considers how the general $\mathcal{H}_\infty$ solution can be used to solve the $\mathcal{H}_\infty$ filtering problem. Section 17.6 considers an alternative approach to the standard $\mathcal{H}_2$ and $\mathcal{H}_\infty$ problems using Youla controller parameterizations, and Section 17.7 gives an 2×2 Hankel-Toeplitz operator interpretations of the $\mathcal{H}_\infty$ solutions presented here and in the last chapter. Finally, the general state feedback $\mathcal{H}_\infty$ control problem and its relations with full information control and differential game problems are discussed in section 17.8 and 17.9.

17.1 General $\mathcal{H}_\infty$ Solutions

Consider the system described by the block diagram



where, as usual, G and K are assumed to be real rational and proper with K constrained to provide internal stability. The controller is said to be admissible if it is real-rational, proper, and stabilizing. Although we are taking everything to be real, the results presented here are still true for the complex case with some obvious modifications. We will again only be interested in characterizing all suboptimal $\mathcal{H}_\infty$ controllers.

The realization of the transfer matrix G is taken to be of the form

$$G(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & 0 \end{array} \right] = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$$

which is compatible with the dimensions $z(t) \in \mathbb{R}^{p_1}$, $y(t) \in \mathbb{R}^{p_2}$, $w(t) \in \mathbb{R}^{m_1}$, $u(t) \in \mathbb{R}^{m_2}$, and the state $x(t) \in \mathbb{R}^n$. The following assumptions are made:

(A1) (A, B_2) is stabilizable and (C_2, A) is detectable;

(A2) $D_{12} = \begin{bmatrix} 0 \\ I \end{bmatrix}$ and $D_{21} = \begin{bmatrix} 0 & I \end{bmatrix}$;

(A3) $\begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ has full column rank for all ω ;

(A4) $\begin{bmatrix} A - j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix}$ has full row rank for all ω .

Assumption (A1) is necessary for the existence of stabilizing controllers. The assumptions in (A2) mean that the penalty on $z = C_1 x + D_{12} u$ includes a nonsingular, normalized penalty on the control u , that the exogenous signal w includes both plant disturbance and sensor noise, and that the sensor noise weighting is normalized and nonsingular. Relaxation of (A2) leads to singular control problems; see Stroorvogel [1990]. For those problems that have D_{12} full column rank and D_{21} full row rank but do not satisfy assumption (A2), a normalizing procedure is given in the next section so that an equivalent new system will satisfy this assumption.

Assumptions (A3) and (A4) are made for a technical reason: together with (A1) they guarantee that the two Hamiltonian matrices in the corresponding $\mathcal{H}_2$ problem belong to $\text{dom}(\text{Ric})$, as we have seen in Chapter 14. It is tempting to suggest that (A3) and (A4) can be dropped, but they are, in some sense, necessary for the methods presented in the last chapter to be applicable. A further discussion of the assumptions and their possible relaxation will be discussed in Section 17.3.

The main result is now stated in terms of the solutions of the X_∞ and Y_∞ Riccati equations together with the “state feedback” and “output injection” matrices F and L .

$$\begin{aligned} R &:= D_{1\bullet}^* D_{1\bullet} - \begin{bmatrix} \gamma^2 I_{m_1} & 0 \\ 0 & 0 \end{bmatrix}, \quad \text{where } D_{1\bullet} := [D_{11} \ D_{12}] \\ \tilde{R} &:= D_{\bullet 1} D_{\bullet 1}^* - \begin{bmatrix} \gamma^2 I_{p_1} & 0 \\ 0 & 0 \end{bmatrix}, \quad \text{where } D_{\bullet 1} := \begin{bmatrix} D_{11} \\ D_{21} \end{bmatrix} \\ H_\infty &:= \begin{bmatrix} A & 0 \\ -C_1^* C_1 & -A^* \end{bmatrix} - \begin{bmatrix} B \\ -C_1^* D_{1\bullet} \end{bmatrix} R^{-1} \begin{bmatrix} D_{1\bullet}^* C_1 & B^* \end{bmatrix} \\ J_\infty &:= \begin{bmatrix} A^* & 0 \\ -B_1 B_1^* & -A \end{bmatrix} - \begin{bmatrix} C^* \\ -B_1 D_{\bullet 1}^* \end{bmatrix} \tilde{R}^{-1} \begin{bmatrix} D_{\bullet 1} B_1^* & C \end{bmatrix} \\ X_\infty &:= \text{Ric}(H_\infty) \quad Y_\infty := \text{Ric}(J_\infty) \end{aligned}$$

$$\begin{aligned} F &:= \begin{bmatrix} F_{1\infty} \\ F_{2\infty} \end{bmatrix} := -R^{-1} [D_{1\bullet}^* C_1 + B^* X_\infty] \\ L &:= \begin{bmatrix} L_{1\infty} & L_{2\infty} \end{bmatrix} := -[B_1 D_{\bullet 1}^* + Y_\infty C^*] \tilde{R}^{-1} \end{aligned}$$

Partition D , $F_{1\infty}$, and $L_{1\infty}$ are as follows:

$$\left[\begin{array}{c|c} & F' \\ \hline L' & D \end{array} \right] = \left[\begin{array}{c|ccc} & F_{11\infty}^* & F_{12\infty}^* & F_{2\infty}^* \\ \hline L_{11\infty}^* & D_{1111} & D_{1112} & 0 \\ L_{12\infty}^* & D_{1121} & D_{1122} & I \\ L_{2\infty}^* & 0 & I & 0 \end{array} \right].$$

Remark 17.1 In the above matrix partitioning, some matrices may not exist depending on whether D_{12} or D_{21} is square. This issue will be discussed further later. For the time being, we shall assume all matrices in the partition exist. $\heartsuit$

Theorem 17.1 Suppose G satisfies the assumptions (A1)–(A4).

- (a) There exists an admissible controller $K(s)$ such that $\|\mathcal{F}_\ell(G, K)\|_\infty < \gamma$ (i.e. $\|T_{zw}\|_\infty < \gamma$) if and only if

- (i) $\gamma > \max(\bar{\sigma}[D_{1111}, D_{1112}], \bar{\sigma}[D_{1111}^*, D_{1121}^*]);$
- (ii) $H_\infty \in \text{dom}(\text{Ric})$ with $X_\infty = \text{Ric}(H_\infty) \geq 0;$
- (iii) $J_\infty \in \text{dom}(\text{Ric})$ with $Y_\infty = \text{Ric}(J_\infty) \geq 0;$
- (iv) $\rho(X_\infty Y_\infty) < \gamma^2.$

(b) Given that the conditions of part (a) are satisfied, then all rational internally stabilizing controllers $K(s)$ satisfying $\|\mathcal{F}_\ell(G, K)\|_\infty < \gamma$ are given by

$$K = \mathcal{F}_\ell(M_\infty, Q) \quad \text{for arbitrary } Q \in \mathcal{RH}_\infty \quad \text{such that} \quad \|Q\|_\infty < \gamma$$

where

$$M_\infty = \left[\begin{array}{c|cc} \hat{A} & \hat{B}_1 & \hat{B}_2 \\ \hline \hat{C}_1 & \hat{D}_{11} & \hat{D}_{12} \\ \hat{C}_2 & \hat{D}_{21} & 0 \end{array} \right]$$

$$\hat{D}_{11} = -D_{1121}D_{1111}^*(\gamma^2 I - D_{1111}D_{1111}^*)^{-1}D_{1112} - D_{1122},$$

$\hat{D}_{12} \in \mathbb{R}^{m_2 \times m_2}$ and $\hat{D}_{21} \in \mathbb{R}^{p_2 \times p_2}$ are any matrices (e.g. Cholesky factors) satisfying

$$\hat{D}_{12}\hat{D}_{12}^* = I - D_{1121}(\gamma^2 I - D_{1111}^*D_{1111})^{-1}D_{1121}^*,$$

$$\hat{D}_{21}^*\hat{D}_{21} = I - D_{1112}^*(\gamma^2 I - D_{1111}D_{1111}^*)^{-1}D_{1112},$$

and

$$\begin{aligned} \hat{B}_2 &= Z_\infty(B_2 + L_{12\infty})\hat{D}_{12}, \\ \hat{C}_2 &= -\hat{D}_{21}(C_2 + F_{12\infty}), \\ \hat{B}_1 &= -Z_\infty L_{2\infty} + \hat{B}_2\hat{D}_{12}^{-1}\hat{D}_{11}, \\ \hat{C}_1 &= F_{2\infty} + \hat{D}_{11}\hat{D}_{21}^{-1}\hat{C}_2, \\ \hat{A} &= A + BF + \hat{B}_1\hat{D}_{21}^{-1}\hat{C}_2, \end{aligned}$$

where

$$Z_\infty = (I - \gamma^{-2}Y_\infty X_\infty)^{-1}.$$

(Note that if $D_{11} = 0$ then the formulae are considerably simplified.)

Some Special Cases:

Case 1: $D_{12} = I$

In this case

1. in part (a), (i) becomes $\gamma > \bar{\sigma}(D_{1121}).$
2. in part (b)

$$\begin{aligned} \hat{D}_{11} &= -D_{1122} \\ \hat{D}_{12}\hat{D}_{12}^* &= I - \gamma^{-2}D_{1121}D_{1121}^* \\ \hat{D}_{21}^*\hat{D}_{21} &= I. \end{aligned}$$

Case 2: $D_{21} = I$

In this case

1. in part (a), (i) becomes $\gamma > \bar{\sigma}(D_{1112})$.
2. in part (b)

$$\begin{aligned}\hat{D}_{11} &= -D_{1122} \\ \hat{D}_{12}\hat{D}_{12}^* &= I \\ \hat{D}_{21}^*\hat{D}_{21} &= I - \gamma^{-2}D_{1112}^*D_{1112}.\end{aligned}$$

Case 3: $D_{12} = I$ & $D_{21} = I$

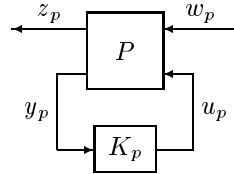
In this case

1. in part (a), (i) drops out.
2. in part (b)

$$\begin{aligned}\hat{D}_{11} &= -D_{1122} \\ \hat{D}_{12}\hat{D}_{12}^* &= I \\ \hat{D}_{21}^*\hat{D}_{21} &= I.\end{aligned}$$

17.2 Loop Shifting

Let a given problem have the following diagram where $z_p(t) \in \mathbb{R}^{p_1}$, $y_p(t) \in \mathbb{R}^{p_2}$, $w_p(t) \in \mathbb{R}^{m_1}$, and $u_p(t) \in \mathbb{R}^{m_2}$:



The plant P has the following state space realization with D_{p12} full column rank and D_{p21} full row rank:

$$P(s) = \left[\begin{array}{c|cc} A_p & B_{p1} & B_{p2} \\ \hline C_{p1} & D_{p11} & D_{p12} \\ C_{p2} & D_{p21} & D_{p22} \end{array} \right]$$

The objective is to find all rational proper controllers $K_p(s)$ which stabilize P and $\|\mathcal{F}_\ell(P, K_p)\|_\infty < \gamma$. To solve this problem we first transfer it to the standard one treated in the last section. *Note that the following procedure can also be applied to the $\mathcal{H}_2$ problem (except the procedure for the case $D_{11} \neq 0$).*

Normalize D_{12} and D_{21}

Perform singular value decompositions to obtain the matrix factorizations

$$D_{p12} = U_p \begin{bmatrix} 0 \\ I \end{bmatrix} R_p, \quad D_{p21} = \tilde{R}_p \begin{bmatrix} 0 & I \end{bmatrix} \tilde{U}_p$$

such that U_p and $\tilde{U}_p$ are square and unitary. Now let

$$z_p = U_p z, \quad w_p = \tilde{U}_p^* w, \quad y_p = \tilde{R}_p y, \quad u_p = R_p u$$

and

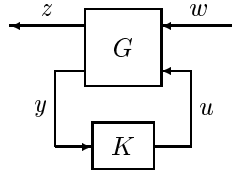
$$K(s) = R_p K_p(s) \tilde{R}_p$$

$$\begin{aligned} G(s) &= \begin{bmatrix} U_p^* & 0 \\ 0 & \tilde{R}_p^{-1} \end{bmatrix} P(s) \begin{bmatrix} \tilde{U}_p^* & 0 \\ 0 & R_p^{-1} \end{bmatrix} \\ &= \left[\begin{array}{c|cc} A_p & B_{p1} \tilde{U}_p^* & B_{p2} R_p^{-1} \\ \hline U_p^* C_{p1} & U_p^* D_{p11} \tilde{U}_p^* & U_p^* D_{p12} R_p^{-1} \\ \tilde{R}_p^{-1} C_{p2} & \tilde{R}_p^{-1} D_{p21} \tilde{U}_p^* & \tilde{R}_p^{-1} D_{p22} R_p^{-1} \end{array} \right] \\ &=: \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right] = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]. \end{aligned}$$

Then

$$D_{12} = \begin{bmatrix} 0 \\ I \end{bmatrix}, \quad D_{21} = \begin{bmatrix} 0 & I \end{bmatrix},$$

and the new system is shown below:



Furthermore, $\|\mathcal{F}_\ell(P, K_p)\|_\alpha = \|U_p \mathcal{F}_\ell(G, K) \tilde{U}_p\|_\alpha = \|\mathcal{F}_\ell(G, K)\|_\alpha$ for $\alpha = 2$ or ∞ since U_p and $\tilde{U}_p$ are unitary.

Remove the Assumption $D_{22} = 0$

Suppose $K(s)$ is a controller for G with D_{22} set to zero. Then the controller for $D_{22} \neq 0$ is $K(I + D_{22}K)^{-1}$. Hence there is no loss of generality in assuming $D_{22} = 0$.

Remove the Assumption $D_{11} = 0$

We can even assume that $D_{11} = 0$. In fact, Theorem 17.1 can be shown by first transforming the general problem to the standard problem considered in the last chapter using Theorem 16.2. This transformation is called *loop-shifting*. Before we go into the detailed description of the transformation, let us first consider a simple unitary matrix dilation problem.

Lemma 17.2 *Suppose D is a constant matrix such that $\|D\| < 1$; then*

$$N = \begin{bmatrix} -D & (I - DD^*)^{1/2} \\ (I - D^*D)^{1/2} & D^* \end{bmatrix}$$

*is a unitary matrix, i.e., $N^*N = I$.*

This result can be easily verified, and the matrix N is called a unitary dilation of D^* (of course, there are many other ways to dilate a matrix to a unitary matrix).

Consider again the feedback system shown at the beginning of this chapter; without loss of generality, we shall assume that the system G has the following realization:

$$G = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & \begin{bmatrix} D_{1111} & D_{1112} \\ D_{1121} & D_{1122} \end{bmatrix} & \begin{bmatrix} 0 \\ I \end{bmatrix} \\ C_2 & \begin{bmatrix} 0 & I \end{bmatrix} & 0 \end{array} \right]$$

with

$$D_{12} = \begin{bmatrix} 0 \\ I \end{bmatrix}, \quad D_{21} = \begin{bmatrix} 0 & I \end{bmatrix}.$$

Suppose there is a stabilizing controller K such that $\|\mathcal{F}_\ell(G, K)\|_\infty < 1$. (Suppose we have normalized γ to 1). In the following, we will show how to construct a new system transfer matrix $M(s)$ and a new controller $\tilde{K}$ such that the D_{11} matrix for $M(s)$ is zero, and, furthermore, $\|\mathcal{F}_\ell(G, K)\|_\infty < 1$ if and only if $\|\mathcal{F}_\ell(M, \tilde{K})\|_\infty < 1$. To begin with, note that $\|\mathcal{F}_\ell(G, K)(\infty)\| < 1$ by the assumption

$$\|\mathcal{F}_\ell(G, K)(\infty)\| = \left\| \begin{bmatrix} D_{1111} & D_{1112} \\ D_{1121} & D_{1122} + K(\infty) \end{bmatrix} \right\|.$$

Let

$$D_\infty \in \left\{ X : \left\| \begin{bmatrix} D_{1111} & D_{1112} \\ D_{1121} & D_{1122} + X \end{bmatrix} \right\| < 1 \right\}.$$

For example, let $D_\infty = -D_{1122} - D_{1121}(I - D_{1111}^*D_{1111})^{-1}D_{1111}^*D_{1112}$ and define

$$\tilde{D}_{11} := \begin{bmatrix} D_{1111} & D_{1112} \\ D_{1121} & D_{1122} + D_\infty \end{bmatrix};$$

then $\|\tilde{D}_{11}\| < 1$. Let

$$K(s) = \tilde{K}(s) + D_\infty$$

to get

$$\mathcal{F}_\ell(G, K) = \mathcal{F}_\ell(G, \tilde{K} + D_\infty) = \mathcal{F}_\ell(\tilde{G}, \tilde{K})$$

where

$$\begin{aligned} \tilde{G} &= \left[\begin{array}{c|cc} A + B_2 D_\infty C_2 & B_1 + B_2 D_\infty D_{21} & B_2 \\ \hline C_1 + D_{12} D_\infty C_2 & \begin{bmatrix} D_{1111} & D_{1112} \\ D_{1121} & D_{1122} + D_\infty \end{bmatrix} & \begin{bmatrix} 0 \\ I \end{bmatrix} \\ \hline C_2 & \begin{bmatrix} 0 & I \end{bmatrix} & 0 \end{array} \right] \\ &=: \left[\begin{array}{c|cc} \tilde{A} & \tilde{B}_1 & \tilde{B}_2 \\ \hline \tilde{C}_1 & \tilde{D}_{11} & \tilde{D}_{12} \\ \hline \tilde{C}_2 & \tilde{D}_{21} & 0 \end{array} \right]. \end{aligned}$$

Let

$$N = \begin{bmatrix} -\tilde{D}_{11} & (I - \tilde{D}_{11} \tilde{D}_{11}^*)^{1/2} \\ (I - \tilde{D}_{11}^* \tilde{D}_{11})^{1/2} & \tilde{D}_{11}^* \end{bmatrix}$$

and then $N^* N = I$. Furthermore, by Theorem 16.2, we have that K stabilizes G and that $\|\mathcal{F}_\ell(G, K)\|_\infty < 1$ if and only if $\mathcal{F}_\ell(G, K)$ internally stabilizes N and

$$\|\mathcal{F}_\ell(N, \mathcal{F}_\ell(G, K))\|_\infty < 1.$$

Note that

$$\mathcal{F}_\ell(N, \mathcal{F}_\ell(G, K)) = \mathcal{F}_\ell(M, \tilde{K})$$

with

$$M(s) = \left[\begin{array}{c|cc} \tilde{A} + \tilde{B}_1 R_1^{-1} \tilde{D}_{11}^* \tilde{C}_1 & \tilde{B}_1 R_1^{-1/2} & \tilde{B}_2 + \tilde{B}_1 R_1^{-1} \tilde{D}_{11}^* \tilde{D}_{12} \\ \hline \tilde{R}_1^{-1/2} \tilde{C}_1 & 0 & \tilde{R}_1^{-1/2} \tilde{D}_{12} \\ \hline \tilde{C}_2 + \tilde{D}_{21} R_1^{-1} \tilde{D}_{11}^* \tilde{C}_1 & \tilde{D}_{21} R_1^{-1/2} & \tilde{D}_{21} R_1^{-1} \tilde{D}_{11}^* \tilde{D}_{12} \end{array} \right]$$

where $R_1 = I - \tilde{D}_{11}^* \tilde{D}_{11}$ and $\tilde{R}_1 = I - \tilde{D}_{11} \tilde{D}_{11}^*$. In summary, we have the following lemma.

Lemma 17.3 *There is a controller K that internally stabilizes G and $\|\mathcal{F}_\ell(G, K)\|_\infty < 1$ if and only if there is a $\tilde{K}$ that stabilizes M and $\|\mathcal{F}_\ell(M, \tilde{K})\|_\infty < 1$.*

Corollary 17.4 *Let $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty$. Then $\|G(s)\|_\infty < 1$ if and only if $\|D\| < 1$,*

$$M(s) = \left[\begin{array}{c|c} A + B(I - D^* D)^{-1} D^* C & B(I - D^* D)^{-1/2} \\ \hline (I - D D^*)^{-1/2} C & 0 \end{array} \right] \in \mathcal{RH}_\infty,$$

and $\|M(s)\|_\infty < 1$.

17.3 Relaxing Assumptions

In this section, we indicate how the results of section 17.1 can be extended to more general cases.

Relaxing A3 and A4

Suppose that

$$G = \left[\begin{array}{c|cc} 0 & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array} \right]$$

which violates both A3 and A4 and corresponds to the robust stabilization of an integrator. If the controller $u = -\epsilon x$ where $\epsilon > 0$ is used, then

$$T_{zw} = \frac{-\epsilon s}{s + \epsilon}, \quad \text{with } \|T_{zw}\|_\infty = \epsilon.$$

Hence the norm can be made arbitrarily small as $\epsilon \rightarrow 0$, but $\epsilon = 0$ is not admissible since it is not stabilizing. This may be thought of as a case where the $\mathcal{H}_\infty$ -optimum is not achieved on the set of admissible controllers. Of course, for this system, $\mathcal{H}_\infty$ optimal control is a silly problem, although the suboptimal case is not obviously so.

If one simply drops the requirement that controllers be admissible and removes assumptions A3 and A4, then the formulae presented above will yield $u = 0$ for both the optimal controller and the suboptimal controller with $\Phi = 0$. This illustrates that assumptions A3 and A4 are necessary for the techniques used in the last chapter to be directly applicable. An alternative is to develop a theory which maintains the same notion of admissibility but relaxes A3 and A4. The easiest way to do this would be to pursue the suboptimal case introducing ϵ perturbations so that A3 and A4 are satisfied.

Relaxing A1

If assumption A1 is violated, then it is obvious that no admissible controllers exist. Suppose A1 is relaxed to allow unstabilizable and/or undetectable modes on the $j\omega$ axis and internal stability is also relaxed to also allow closed-loop $j\omega$ axis poles, but A2-A4 is still satisfied. It can be easily shown that under these conditions the closed-loop $\mathcal{H}_\infty$ norm cannot be made finite and, in particular, that the unstabilizable and/or undetectable modes on the $j\omega$ axis must show up as poles in the closed-loop system, see Lemma 16.1.

Violating A1 *and* either or both of A3 and A4

Sensible control problems can be posed which violate A1 *and* either or both of A3 and A4. For example, cases when A has modes at $s = 0$ which are unstabilizable through B_2 and/or undetectable through C_2 arise when an integrator is included in a weight on a disturbance input or an error term. In these cases, either A3 or A4 are also violated, or the closed-loop $\mathcal{H}_\infty$ norm cannot be made finite. In many applications, such problems can be reformulated so that the integrator occurs inside the loop (essentially using the internal model principle) and is hence detectable and stabilizable. We will show this process in the next section.

An alternative approach to such problems which could potentially avoid the problem reformulation would be to pursue the techniques in the last chapter but to relax internal stability to the requirement that all closed-loop modes be in the closed left half plane. Clearly, to have finite $\mathcal{H}_\infty$ norms, these closed-loop modes could not appear as poles in T_{zw} . The formulae given in this chapter will often yield controllers compatible with these assumptions. The user would then have to decide whether closed-loop poles on the imaginary axis were due to weights and hence acceptable or due to the problem being poorly posed, as in the above example.

A third alternative is to again introduce ϵ perturbations so that A1, A3, and A4 are satisfied. Roughly speaking, this would produce sensible answers for sensible problems, but the behavior as $\epsilon \rightarrow 0$ could be problematic.

Relaxing A2

In the cases that either D_{12} is not full column rank or that D_{21} is not full row rank, improper controllers can give a bounded $\mathcal{H}_\infty$ -norm for T_{zw} , although the controllers will not be admissible by our definition. Such singular filtering and control problems have been well-studied in $\mathcal{H}_2$ theory and many of the same techniques go over to the $\mathcal{H}_\infty$ -case (e.g. Willems [1981], Willems, Kitapci, and Silverman [1986], and Hautus and Silverman [1983]). In particular, the structure algorithm of Silverman [1969] could be used to make the terms D_{12} and D_{21} full rank by the introduction of suitable differentiators in the controller. A complete solution to the singular problem can be found in [Stroorvogel, 1990].

17.4 $\mathcal{H}_2$ and $\mathcal{H}_\infty$ Integral Control

It is interesting to note that the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ design frameworks do not in general produce integral control. In this section we show how to introduce integral control into the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ design framework through a simple disturbance rejection problem. We consider a feedback system shown in Figure 17.1. We shall assume that the frequency contents of the disturbance w are effectively modeled by the weighting $W_d \in \mathcal{RH}_\infty$ and the constraints on control signal are limited by an appropriate choice of $W_u \in \mathcal{RH}_\infty$. In order to let the output y track the reference signal r , we require K contain an integral,

i.e., $K(s)$ has a pole at $s = 0$. (In general, K is required to have poles on the imaginary axis.)

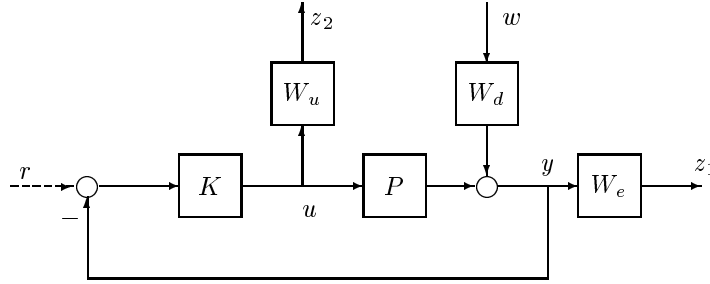


Figure 17.1: A Simple Disturbance Rejection Problem

There are several ways to achieve the integral design. One approach is to introduce an integral in the performance weight W_e . Then the transfer function between w and z_1 is given by

$$z_1 = W_e(I + PK)^{-1}W_d w.$$

Now if the resulting controller K stabilizes the plant P and makes the norm (2-norm or ∞ -norm) between w and z_1 finite, then K must have a pole at $s = 0$ which is the zero of the sensitivity function (Assuming W_d has no zeros at $s = 0$). (This follows from the well-known internal model principle.) The problem with this approach is that the $\mathcal{H}_2$ (or $\mathcal{H}_\infty$) control theory presented in this chapter and in the previous chapters can not be applied to this problem formulation directly because the pole $s = 0$ of W_e becomes an uncontrollable pole of the feedback system and the very first assumption for the $\mathcal{H}_2$ (or $\mathcal{H}_\infty$) theory is violated.

However, the obstacles can be overcome by appropriately reformulating the problem. Suppose W_e can be factorized as follows

$$W_e = \tilde{W}_e(s)M(s)$$

where $M(s)$ is proper, containing all the imaginary axis poles of W_e , and $M^{-1}(s) \in \mathcal{RH}_\infty$, $\tilde{W}_e(s)$ is stable and minimum phase. Now suppose there exists a controller $K(s)$ which contains the same imaginary axis poles that achieves the performance. Then without loss of generality, K can be factorized as

$$K(s) = -\hat{K}(s)M(s)$$

where there is no unstable pole/zero cancellation in forming the product $\hat{K}(s)M(s)$. Now the problem can be reformulated as in Figure 17.2. Figure 17.2 can be put in the general LFT framework as in Figure 17.3.

Let $\tilde{W}_e$, W_u , W_d , M , and P have the following stabilizable and detectable state space

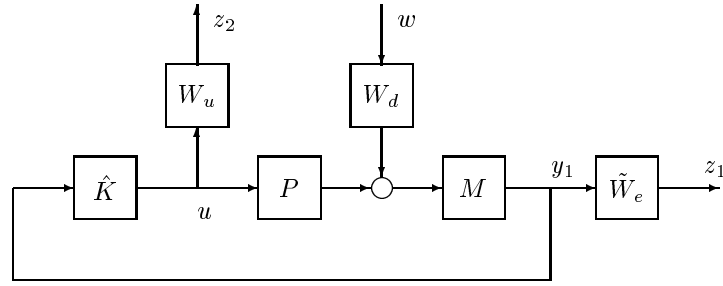


Figure 17.2: Disturbance Rejection with Imaginary Axis Poles

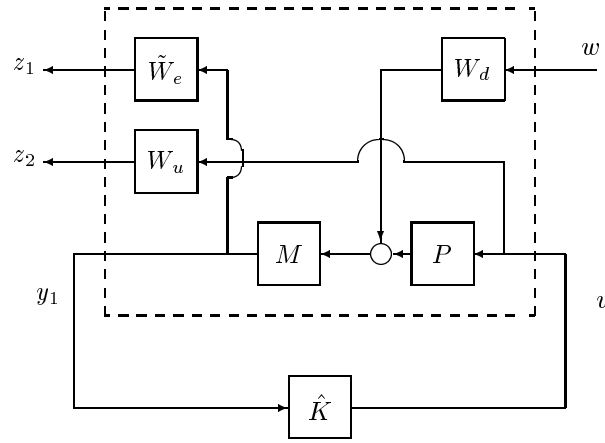


Figure 17.3: LFT Framework for the Disturbance Rejection Problem

realizations:

$$\tilde{W}_e = \left[\begin{array}{c|c} A_e & B_e \\ \hline C_e & D_e \end{array} \right], \quad W_u = \left[\begin{array}{c|c} A_u & B_u \\ \hline C_u & D_u \end{array} \right], \quad W_d = \left[\begin{array}{c|c} A_d & B_d \\ \hline C_d & D_d \end{array} \right]$$

$$M = \left[\begin{array}{c|c} A_m & B_m \\ \hline C_m & D_m \end{array} \right], \quad P = \left[\begin{array}{c|c} A_p & B_p \\ \hline C_p & D_p \end{array} \right].$$

Then a realization for the generalized system is given by

$$G(s) = \left[\begin{array}{c} \left[\begin{array}{c|c} \tilde{W}_e M W_d \\ \hline 0 \end{array} \right] \\ \left[\begin{array}{c|c} \tilde{W}_e M P \\ \hline W_u \end{array} \right] \\ \left[\begin{array}{c|c} M W_d \\ \hline M P \end{array} \right] \end{array} \right]$$

$$= \left[\begin{array}{ccccc|cc} A_e & 0 & B_e C_m & B_e D_m C_d & B_e D_m C_p & B_e D_m D_d & B_e D_m D_p \\ 0 & A_u & 0 & 0 & 0 & 0 & B_u \\ 0 & 0 & A_m & B_m C_d & B_m C_p & B_m D_d & B_m D_p \\ 0 & 0 & 0 & A_d & 0 & B_d & 0 \\ 0 & 0 & 0 & 0 & A_p & 0 & B_p \\ \hline C_e & 0 & D_e C_m & D_e D_m C_d & D_e D_m C_p & D_e D_m D_d & D_e D_m D_p \\ 0 & C_u & 0 & 0 & 0 & 0 & D_u \\ 0 & 0 & C_m & D_m C_d & D_m C_p & D_m D_d & D_m D_p \end{array} \right].$$

We shall illustrate the above design through a simple numerical example. Let

$$P = \frac{s-2}{(s+1)(s-3)} = \left[\begin{array}{cc|c} 0 & 1 & 0 \\ 3 & 2 & 1 \\ \hline -2 & 1 & 0 \end{array} \right], \quad W_d = 1,$$

$$W_u = \frac{s+10}{s+100} = \left[\begin{array}{cc|c} -100 & -90 & \\ \hline 1 & 1 & \end{array} \right], \quad W_e = \frac{1}{s}.$$

Then we can choose without loss of generality that

$$M = \frac{s+\alpha}{s}, \quad \tilde{W}_e = \frac{1}{s+\alpha}, \quad \alpha > 0.$$

This gives the following generalized system

$$G(s) = \left[\begin{array}{ccccc|cc} -\alpha & 0 & 1 & -2 & 1 & 1 & 0 \\ 0 & -100 & 0 & 0 & 0 & 0 & -90 \\ 0 & 0 & 0 & -2\alpha & \alpha & \alpha & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 3 & 2 & 0 & 1 \\ \hline 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -2 & 1 & 1 & 0 \end{array} \right]$$

and suboptimal $\mathcal{H}_\infty$ controller $\hat{K}_\infty$ for each α can be computed easily as

$$\hat{K}_\infty = \frac{-2060381.4(s+1)(s+\alpha)(s+100)(s-0.1557)}{(s+\alpha)^2(s+32.17)(s+262343)(s-19.89)}$$

which gives the closed-loop ∞ norm 7.854. Hence the controller $K_\infty = -\hat{K}_\infty(s)M(s)$ is given by

$$K_\infty(s) = \frac{2060381.4(s+1)(s+100)(s-0.1557)}{s(s+32.17)(s+262343)(s-19.89)} \approx \frac{7.85(s+1)(s+100)(s-0.1557)}{s(s+32.17)(s-19.89)}$$

which is independent of α as expected. Similarly, we can solve an optimal $\mathcal{H}_2$ controller

$$\hat{K}_2 = \frac{-43.487(s+1)(s+\alpha)(s+100)(s-0.069)}{(s+\alpha)^2(s^2+30.94s+411.81)(s-7.964)}$$

and

$$K_2(s) = -\hat{K}_2(s)M(s) = \frac{43.487(s+1)(s+100)(s-0.069)}{s(s^2+30.94s+411.81)(s-7.964)}.$$

An approximate integral control can also be achieved without going through the above process by letting

$$W_e = \tilde{W}_e = \frac{1}{s+\epsilon}, \quad M(s) = 1$$

for a sufficiently small $\epsilon > 0$. For example, a controller for $\epsilon = 0.001$ is given by

$$K_\infty = \frac{316880(s+1)(s+100)(s-0.1545)}{(s+0.001)(s+32)(s+40370)(s-20)} \approx \frac{7.85(s+1)(s+100)(s-0.1545)}{s(s+32)(s-20)}$$

which gives the closed-loop $\mathcal{H}_\infty$ norm of 7.85. Similarly, an approximate $\mathcal{H}_2$ integral controller is obtained as

$$K_2 = \frac{43.47(s+1)(s+100)(s-0.0679)}{(s+0.001)(s^2+30.93s+411.7)(s-7.9718)}.$$

17.5 $\mathcal{H}_\infty$ Filtering

In this section we show how the filtering problem can be solved using the $\mathcal{H}_\infty$ theory developed early. Suppose a dynamic system is described by the following equations

$$\dot{x} = Ax + B_1 w(t), \quad x(0) = 0 \quad (17.1)$$

$$y = C_2 x + D_{21} w(t) \quad (17.2)$$

$$z = C_1 x + D_{11} w(t) \quad (17.3)$$

The filtering problem is to find an estimate $\hat{z}$ of z in some sense using the measurement of y . The restriction on the filtering problem is that the filter has to be causal so that it can be realized, i.e., $\hat{z}$ has to be generated by a causal system acting on the measurements. We will further restrict our filter to be *unbiased*, i.e., given $T > 0$ the estimate $\hat{z}(t) = 0 \quad \forall t \in [0, T]$ if $y(t) = 0, \quad \forall t \in [0, T]$. Now we can state our $\mathcal{H}_\infty$ filtering problem.

$\mathcal{H}_\infty$ Filtering: Given a $\gamma > 0$, find a causal filter $F(s) \in \mathcal{RH}_\infty$ if it exists such that

$$J := \sup_{w \in \mathcal{L}_2[0, \infty)} \frac{\|z - \hat{z}\|_2^2}{\|w\|_2^2} < \gamma^2$$

with $\hat{z} = F(s)y$.

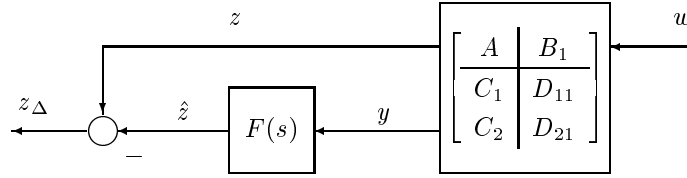
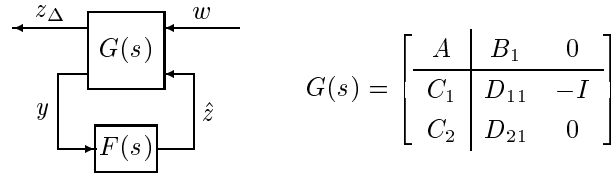


Figure 17.4: Filtering Problem Formulation

A diagram for the filtering problem is shown in Figure 17.4.

The above filtering problem can also be formulated in an LFT framework: given a system shown below



find a filter $F(s) \in \mathcal{RH}_\infty$ such that

$$\sup_{w \in \mathcal{L}_2} \frac{\|z_\Delta\|_2^2}{\|w\|_2^2} < \gamma^2. \quad (17.4)$$

Hence the filtering problem can be regarded as a special $\mathcal{H}_\infty$ problem. However, comparing with control problems there is no internal stability requirement in the filtering problem. Hence the solution to the above filtering problem can be obtained from the $\mathcal{H}_\infty$ solution in the previous sections by setting $B_2 = 0$ and dropping the internal stability requirement.

Theorem 17.5 Suppose (C_2, A) is detectable and

$$\begin{bmatrix} A - j\omega I & B_1 \\ C_2 & D_{21} \end{bmatrix}$$

has full row rank for all ω . Let D_{21} be normalized and D_{11} partitioned conformably as

$$\begin{bmatrix} D_{11} \\ D_{21} \end{bmatrix} = \begin{bmatrix} D_{111} & D_{112} \\ 0 & I \end{bmatrix}.$$

Then there exists a causal filter $F(s) \in \mathcal{RH}_\infty$ such that $J < \gamma^2$ if and only if $\overline{\sigma}(D_{111}) < \gamma$ and $J_\infty \in \text{dom}(\text{Ric})$ with $Y_\infty = \text{Ric}(J_\infty) \geq 0$ where

$$\tilde{R} := \begin{bmatrix} D_{11} \\ D_{21} \end{bmatrix} \begin{bmatrix} D_{11} \\ D_{21} \end{bmatrix}^* - \begin{bmatrix} \gamma^2 I & 0 \\ 0 & 0 \end{bmatrix}$$

$$J_\infty := \begin{bmatrix} A^* & 0 \\ -B_1 B_1^* & -A \end{bmatrix} - \begin{bmatrix} C_1^* & C_2^* \\ -B_1 D_{11}^* & -B_1 D_{21}^* \end{bmatrix} \tilde{R}^{-1} \begin{bmatrix} D_{11} B_1^* & C_1 \\ D_{21} B_1^* & C_2 \end{bmatrix}.$$

Moreover, if the above conditions are satisfied, then a rational causal filter $F(s)$ satisfying $J < \gamma^2$ is given by

$$\hat{z} = F(s)y = \left[\begin{array}{c|c} A + L_{2\infty}C_2 + L_{1\infty}D_{112}C_2 & -L_{2\infty} - L_{1\infty}D_{112} \\ \hline C_1 - D_{112}C_2 & D_{112} \end{array} \right] y$$

where

$$\begin{bmatrix} L_{1\infty} & L_{2\infty} \end{bmatrix} := - \begin{bmatrix} B_1 D_{11}^* + Y_\infty C_1^* & B_1 D_{21}^* + Y_\infty C_2^* \end{bmatrix} \tilde{R}^{-1}.$$

In the case where $D_{11} = 0$ and $B_1 D_{21}^* = 0$ the filter becomes much simpler:

$$\hat{z} = \left[\begin{array}{c|c} A - Y_\infty C_2^* C_2 & Y_\infty C_2^* \\ \hline C_1 & 0 \end{array} \right] y$$

where Y_∞ is the stabilizing solution to

$$Y_\infty A^* + A Y_\infty + Y_\infty (\gamma^{-2} C_1^* C_1 - C_2^* C_2) Y_\infty + B_1 B_1^* = 0.$$

17.6 Youla Parameterization Approach*

In this section, we shall briefly outline an alternative approach to solving the standard $\mathcal{H}_2$ and $\mathcal{H}_\infty$ control problems using the Q parameterization (Youla parameterization) approach. This approach was the main focus in the early 1980's and is still very useful in solving many interesting problems. We will see that this approach may suggest additional interpretations for the results presented in the last chapter and applies to both $\mathcal{H}_2$ and $\mathcal{H}_\infty$ problems. The $\mathcal{H}_2$ problem is very simple and involves a projection. While the $\mathcal{H}_\infty$ problem is much more difficult, it bears some similarities to the constant matrix dilation problem but with the restriction of internal stability. Nevertheless, we have built enough tools to give a fairly complete solution using this Q parameterization approach.

Assume again that the G has the realization

$$G = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right]$$

with the same assumptions as before. But for convenience, we will allow the assumption (A2) to be relaxed to the following

(A2') D_{12} is full column rank with $\begin{bmatrix} D_{12} & D_{\perp} \end{bmatrix}$ unitary, and D_{21} is full row rank with $\begin{bmatrix} D_{21} \\ \tilde{D}_{\perp} \end{bmatrix}$ unitary.

Next, we will outline the steps required to solve the $\mathcal{H}_2$ and $\mathcal{H}_{\infty}$ control problems. Because of the similarity between the $\mathcal{H}_2$ and $\mathcal{H}_{\infty}$ problems, they are developed in parallel below.

Parameterization: Recall that all controllers stabilizing the plant G can be expressed as

$$K = \mathcal{F}_l(M_2, Q), \quad I + D_{22}Q(\infty) \text{ invertible}$$

with $Q \in \mathcal{RH}_{\infty}$ and

$$M_2 = \left[\begin{array}{c|cc} A + B_2F_2 + L_2C_2 + L_2D_{22}F_2 & -L_2 & B_2 + L_2D_{22} \\ \hline F_2 & 0 & I \\ \hline -(C_2 + D_{22}F_2) & I & -D_{22} \end{array} \right]$$

where

$$\begin{aligned} F_2 &= -(B_2^*X_2 + D_{12}^*C_1) \\ L_2 &= -(Y_2C_2^* + B_1D_{21}^*) \\ X_2 &= \text{Ric} \begin{bmatrix} A - B_2D_{12}^*C_1 & -B_2B_2^* \\ -C_1^*D_{\perp}D_{\perp}^*C_1 & -(A - B_2D_{12}^*C_1)^* \end{bmatrix} \geq 0 \\ Y_2 &= \text{Ric} \begin{bmatrix} (A - B_1D_{21}^*C_2)^* & -C_2^*C_2 \\ -B_1\tilde{D}_{\perp}^*\tilde{D}_{\perp}B_1^* & -(A - B_1D_{21}^*C_2) \end{bmatrix} \geq 0. \end{aligned}$$

Then the closed-loop transfer matrix from w to z can be written as

$$\mathcal{F}_l(G, K) = T_o + UQV, \quad I + D_{22}Q(\infty) \text{ invertible}$$

where

$$\begin{aligned} T_o &= \left[\begin{array}{cc|c} A_{F_2} & -B_2F_2 & B_1 \\ 0 & A_{L_2} & B_{1L_2} \\ \hline C_{1F_2} & -D_{12}F_2 & D_{11} \end{array} \right] \in \mathcal{RH}_{\infty} \\ U &= \left[\begin{array}{c|c} A_{F_2} & B_2 \\ \hline C_{1F_2} & D_{12} \end{array} \right], \quad V = \left[\begin{array}{c|c} A_{L_2} & B_{1L_2} \\ \hline C_2 & D_{21} \end{array} \right] \end{aligned}$$

and

$$\begin{aligned} A_{F_2} &:= A + B_2F_2, \quad A_{L_2} := A + L_2C_2 \\ C_{1F_2} &= C_1 + D_{12}F_2, \quad B_{1L_2} = B_1 + L_2D_{21}. \end{aligned}$$

It is easy to show that U is an inner, $U^{\sim}U = I$, and V is a co-inner, $VV^{\sim} = I$.

Unitary Invariant: There exist $U_\perp$ and $V_\perp$ such that $\begin{bmatrix} U & U_\perp \end{bmatrix}$ and $\begin{bmatrix} V \\ V_\perp \end{bmatrix}$ are square and inner:

$$U_\perp = \left[\begin{array}{c|c} A_{F_2} & -X_2^+ C_1^* D_\perp \\ \hline C_{1F_2} & D_\perp \end{array} \right], \quad V_\perp = \left[\begin{array}{c|c} A_{L_2} & B_{1L_2} \\ \hline -\tilde{D}_\perp B_1^* Y_2^+ & \tilde{D}_\perp \end{array} \right].$$

Since the multiplication of square inner matrices do not change $\mathcal{H}_2$ and $\mathcal{H}_\infty$ norms, we have for $\alpha = 2$ or ∞

$$\begin{aligned} \|\mathcal{F}_\ell(P, K)\|_\alpha &= \|T_o + UQV\|_\alpha \\ &= \left\| \begin{bmatrix} U & U_\perp \end{bmatrix}^\sim (T_o + UQV) \begin{bmatrix} V \\ V_\perp \end{bmatrix} \right\|_\alpha^\sim \\ &= \left\| \begin{bmatrix} R_{11} + Q & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \right\|_\alpha \end{aligned}$$

where

$$R = \begin{bmatrix} U^\sim \\ U_\perp^\sim \end{bmatrix} T_o \begin{bmatrix} V^\sim & V_\perp^\sim \end{bmatrix}$$

with the obvious partitioning. It can be shown through some long and tedious algebra that R is antistable and has the following representation:

$$R = \left[\begin{array}{cc|cc} -A_{F_2}^* & EB_{1L_2}^* & -ED_{21}^* & -E\tilde{D}_\perp^* \\ 0 & -A_{L_2}^* & C_2^* & -Y_2^+ B_1 \tilde{D}_\perp^* \\ \hline B_2^* & F_2 Y_2 - D_{12}^* D_{11} B_{1L_2}^* & D_{12}^* D_{11} D_{21}^* & D_{12}^* D_{11} \tilde{D}_\perp^* \\ -D_\perp^* C_1 X_2^+ & -D_\perp^* D_{11} B_{1L_2}^* & D_\perp^* D_{11} D_{21}^* & D_\perp^* D_{11} \tilde{D}_\perp^* \end{array} \right]$$

where $E := X_2 B_1 + C_{1F_2}^* D_{11}$.

Projection/Dilation: At this point the $\alpha = 2$ and $\alpha = \infty$ cases have to be treated separately.

$\mathcal{H}_2$ case: In this case, we will assume $D_\perp^* D_{11} = 0$ and $D_{11} \tilde{D}_\perp^* = 0$; otherwise, the 2-norm of the closed loop transfer matrix will be unbounded since

$$R(\infty) = \begin{bmatrix} D_{12}^* D_{11} D_{21}^* & D_{12}^* D_{11} \tilde{D}_\perp^* \\ D_\perp^* D_{11} D_{21}^* & D_\perp^* D_{11} \tilde{D}_\perp^* \end{bmatrix}.$$

Now from the definition of 2-norm, the problem can be rewritten as

$$\|\mathcal{F}_\ell(G, K)\|_2^2 = \left\| \begin{bmatrix} R_{11} + Q & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \right\|_2^2$$

$$= \|R_{11} + Q\|_2^2 + \left\| \begin{bmatrix} 0 & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \right\|_2^2.$$

Furthermore,

$$\|R_{11} + Q\|_2^2 = \|R_{11} - D_{12}^* D_{11} D_{21}^*\|_2^2 + \|Q + D_{12}^* D_{11} D_{21}^*\|_2^2$$

since $(R_{11} - D_{12}^* D_{11} D_{21}^*) \in \mathcal{H}_2^\perp$ and $(Q + D_{12}^* D_{11} D_{21}^*) \in \mathcal{H}_\infty$. In fact, $(Q + D_{12}^* D_{11} D_{21}^*)$ has to be in $\mathcal{H}_2$ to guarantee that the 2-norm be finite.

Hence the unique optimal solution is given by a projection to $\mathcal{H}_\infty$:

$$Q_{opt} = [-R_{11}]_+ = -D_{12}^* D_{11} D_{21}^*.$$

The optimal controller is given by

$$K_{opt} = \mathcal{F}_l(M_2, Q_{opt}).$$

In particular, if $D_{11} = 0$ then $K_{opt} = \mathcal{F}_l(M_2, 0)$.

$\mathcal{H}_\infty$ **case:** Recall the definition of the ∞ -norm:

$$\left\| \begin{bmatrix} R_{11} + Q & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \right\|_\infty = \sup_\omega \bar{\sigma} \left(\begin{bmatrix} R_{11} + Q & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \right) (j\omega).$$

Consider the minimization problem with respect to Q frequency by frequency; it becomes a constant matrix dilation problem if no causality restrictions are imposed on Q . The $\mathcal{H}_\infty$ optimal control follows this idea. However, it should be noted that since Q is restricted to be an $\mathcal{RH}_\infty$ matrix, the term $R_{11} + Q$ cannot be made into an arbitrary matrix. Therefore, the problem will be significantly different from the constant matrix case, and, generically,

$$\min_{Q \in \mathcal{H}_\infty} \left\| \begin{bmatrix} R_{11} + Q & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \right\|_\infty > \gamma_0$$

where

$$\gamma_0 := \max \left\{ \left\| \begin{bmatrix} R_{21} & R_{22} \end{bmatrix} \right\|_\infty, \left\| \begin{bmatrix} R_{12} \\ R_{22} \end{bmatrix} \right\|_\infty \right\}.$$

It is convenient to use the characterization in Corollary 2.23. Recall that

$$\left\| \begin{bmatrix} R_{11} + Q & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \right\|_\infty \leq \gamma \quad (< \gamma), \text{ for } \gamma > \gamma_0$$

iff

$$\left\| (I - WW^\sim)^{-1/2} (Q + R_{11} + WR_{22}^\sim Z) (I - Z^\sim Z)^{-1/2} \right\|_\infty \leq \gamma \quad (< \gamma)$$

where

$$\begin{aligned} W &= R_{12}(\gamma^2 I - R_{22}^\sim R_{22})^{-1/2} \\ Z &= (\gamma^2 I - R_{22}^\sim R_{22})^{-1/2} R_{21}. \end{aligned}$$

The key here is to find the spectral factors $(I - WW^\sim)^{1/2}$ and $(I - Z^\sim Z)^{1/2}$ with stable inverses such that if $M = (I - WW^\sim)^{1/2}$ and $N = (I - Z^\sim Z)^{1/2}$, then $M, M^{-1}, N, N^{-1} \in \mathcal{H}_\infty$; $MM^\sim = I - WW^\sim$, and $N^\sim N = (I - Z^\sim Z)$. Now let

$$\hat{Q} := M^{-1}QN^{-1}, \quad G := M^{-1}(R_{11} + WR_{22}^\sim Z)N^{-1};$$

then the problem is reduced to finding $\hat{Q} \in \mathcal{H}_\infty$ such that

$$\|G + \hat{Q}\|_\infty \leq \gamma \quad (< \gamma). \quad (17.5)$$

Note that $Q = M\hat{Q}N \in \mathcal{H}_\infty$ iff $\hat{Q} \in \mathcal{H}_\infty$.

The final step in the $\mathcal{H}_\infty$ problem involves solving (17.5) for $\hat{Q} \in \mathcal{H}_\infty$. This is a standard Nehari problem and is solved in Chapter 8. The optimal control law will be given by

$$K_{opt} = \mathcal{F}_\ell(J, Q_{opt}).$$

17.7 Connections

This section considers the connections between the Youla parameterization approach and the current state space approach discussed in the last chapter. The key is Lemma 15.7 and its connection with the formulae given in the last section.

To see how Lemma 15.7 might be used in the last section to prove Theorems 16.4 and 16.5 or Theorem 17.1, we begin with G having state dimension n . For simplicity, we assume further that $D_{11} = 0$. Then from the last section, we have

$$\begin{aligned} \|T_{zw}\|_\infty &= \left\| R + \begin{bmatrix} Q & 0 \\ 0 & 0 \end{bmatrix} \right\|_\infty \\ &= \left\| \begin{bmatrix} R_{11} + Q & R_{12} \\ R_{21} & R_{22} \end{bmatrix} \right\|_\infty \\ &= \left\| R^\sim + \begin{bmatrix} Q^\sim & 0 \\ 0 & 0 \end{bmatrix} \right\|_\infty \end{aligned}$$

where R has state dimension $2n$. Now define

$$\begin{bmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{bmatrix} := R^\sim.$$

Then

$$\begin{bmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{bmatrix} = \left[\begin{array}{cc|cc} A_{F_2} & 0 & B_2 & -X_2^+ C_1^* D_\perp \\ -B_{1L_2} B_1^* X_2 & A_{L_2} & Y_2 F_2^* & 0 \\ \hline D_{21} B_1^* X_2 & -C_2 & 0 & 0 \\ \tilde{D}_\perp B_1^* X_2 & \tilde{D}_\perp B_1^* Y_2^+ & 0 & 0 \end{array} \right] \in \mathcal{RH}_\infty, \quad (17.6)$$

and the $\mathcal{H}_\infty$ problem becomes to find an antistable $Q^\sim$ such that

$$\left\| \begin{bmatrix} N_{11} + Q^\sim & N_{12} \\ N_{21} & N_{22} \end{bmatrix} \right\|_\infty < \gamma.$$

Let $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ and note that

$$\begin{aligned} \left\| \begin{bmatrix} N_{11} + Q^\sim & N_{12} \\ N_{21} & N_{22} \end{bmatrix} \right\|_\infty^2 &:= \sup_{w \in \mathcal{BL}_2} \left\| \begin{bmatrix} N_{11} + Q^\sim & N_{12} \\ N_{21} & N_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \right\|_2^2 \\ &= \sup_{\{w \in \mathcal{BL}_2\} \cap \{w_1 \in \mathcal{H}_2^\perp\}} \left\| \begin{bmatrix} N_{11} + Q^\sim & N_{12} \\ N_{21} & N_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \right\|_2^2. \end{aligned}$$

The right hand side of the above equation can be written as

$$\begin{aligned} \sup_{\{w \in \mathcal{BL}_2\} \cap \{w_1 \in \mathcal{H}_2^\perp\}} \left\{ \left\| \begin{bmatrix} P_+ & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \right\|_2^2 \right. \\ \left. + \|Q^\sim w_1 + P_-(N_{11}w_1 + N_{12}w_2)\|_2^2 \right\}. \end{aligned}$$

It is clear that the last term can be made zero for an appropriate choice of an antistable $Q^\sim$. Hence the $\mathcal{H}_\infty$ problem has a solution if and only if

$$\sup_{\{w \in \mathcal{BL}_2\} \cap \{w_1 \in \mathcal{H}_2^\perp\}} \left\| \begin{bmatrix} P_+ & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} N_{11} & N_{12} \\ N_{21} & N_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \right\|_2 < \gamma.$$

But this is exactly the same operator as the one in Lemma 15.7. Lemma 15.7 may be applied to derive the solvability conditions and some additional arguments to construct a $Q \in \mathcal{RH}_\infty$ from X and Y such that $\|T_{zw}(Q)\|_\infty < \gamma$. In fact, it turns out that X in Lemma 15.7 for N is exactly W in the FI proof or in the differential game problem.

The final step is to obtain the controller from M_2 and Q . Since M_2 has state dimension n and Q has $2n$, the apparent state dimension of K is $3n$, but some tedious state space manipulations produce cancelations resulting in the n dimensional controller

formulae in Theorems 16.4 and 16.5. This approach is exactly the procedure used in Doyle [1984] and Francis [1987] with Lemma 15.7 used to solve the general distance problem. Although this approach is conceptually straightforward and was, in fact, used to obtain the first proof of the current state space results in this chapter, it seems unnecessarily cumbersome and indirect.

17.8 State Feedback and Differential Game

It has been shown in Chapters 15 and 16 that a (central) suboptimal full information $\mathcal{H}_\infty$ control law is actually a pure state feedback if $D_{11} = 0$. However, this is not true in general if $D_{11} \neq 0$, as will be shown below. Nevertheless, the state feedback $\mathcal{H}_\infty$ control is itself a very interesting problem and deserves special treatment. This section and the section to follow are devoted to the study of this state feedback $\mathcal{H}_\infty$ control problem and its connections with full information control and differential game.

Consider a dynamical system

$$\dot{x} = Ax + B_1 w + B_2 u \quad (17.7)$$

$$z = C_1 x + D_{11} w + D_{12} u \quad (17.8)$$

where $z(t) \in \mathbb{R}^{p_1}$, $y(t) \in \mathbb{R}^{p_2}$, $w(t) \in \mathbb{R}^{m_1}$, $u(t) \in \mathbb{R}^{m_2}$, and $x(t) \in \mathbb{R}^n$. The following assumptions are made:

(AS1) (A, B_2) is stabilizable;

(AS2) There is a matrix $D_\perp$ such that $\begin{bmatrix} D_{12} & D_\perp \end{bmatrix}$ is unitary;

(AS3) $\begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix}$ has full column rank for all ω .

We are interested in the following two related quadratic min-max problems: given $\gamma > 0$, check if

$$\sup_{w \in B\mathcal{L}_2[0, \infty)} \min_{u \in \mathcal{L}_2[0, \infty)} \|z\|_2 < \gamma$$

and

$$\min_{u \in \mathcal{L}_2[0, \infty)} \sup_{w \in B\mathcal{L}_2[0, \infty)} \|z\|_2 < \gamma.$$

The first problem can be regarded as a full information control problem since the control signal u can be a function of the disturbance w and the system state x . On the other hand, the optimal control signal in the second problem cannot depend on the disturbance w (the worst disturbance w can be a function of u and x). In fact, it will be shown that the control signal in the latter problem depends only on the system state; hence, it is equivalent to a state feedback control.

Theorem 17.6 *Let $\gamma > 0$ be given and define*

$$R := D_{1\bullet}^* D_{1\bullet} - \begin{bmatrix} \gamma^2 I_{m_1} & 0 \\ 0 & 0 \end{bmatrix}, \quad \text{where } D_{1\bullet} := [D_{11} \ D_{12}]$$

$$H_\infty := \begin{bmatrix} A & 0 \\ -C_1^* C_1 & -A^* \end{bmatrix} - \begin{bmatrix} B \\ -C_1^* D_{1\bullet} \end{bmatrix} R^{-1} \begin{bmatrix} D_{1\bullet}^* C_1 & B^* \end{bmatrix}$$

$$\text{where } B := \begin{bmatrix} B_1 & B_2 \end{bmatrix}.$$

(a) $\sup_{w \in B\mathcal{L}_2[0, \infty)} \min_{u \in \mathcal{L}_2[0, \infty)} \|z\|_2 < \gamma$ if and only if

$$\bar{\sigma}(D_{1\bullet}^* D_{11}) < \gamma, \quad H_\infty \in \text{dom}(\text{Ric}), \quad X_\infty = \text{Ric}(H_\infty) \geq 0.$$

Moreover, an optimal u is given by

$$u = -D_{12}^* D_{11} w + \begin{bmatrix} D_{12}^* D_{11} & I \end{bmatrix} F x,$$

and a worst w_{fworst} is given by

$$w_{\text{fworst}} = F_{1\infty} x$$

where

$$F := \begin{bmatrix} F_{1\infty} \\ F_{2\infty} \end{bmatrix} := -R^{-1} [D_{1\bullet}^* C_1 + B^* X_\infty].$$

(b) $\min_{u \in \mathcal{L}_2[0, \infty)} \sup_{w \in B\mathcal{L}_2[0, \infty)} \|z\|_2 < \gamma$ if and only if

$$\bar{\sigma}(D_{11}) < \gamma, \quad H_\infty \in \text{dom}(\text{Ric}), \quad X_\infty = \text{Ric}(H_\infty) \geq 0.$$

Moreover, an optimal u is given by

$$u = F_{2\infty} x = -D_{12}^* D_{11} w_{\text{fworst}} + \begin{bmatrix} D_{12}^* D_{11} & I \end{bmatrix} F x,$$

and the worst w_{sfworst} is given by

$$w_{\text{sfworst}} = (\gamma^2 I - D_{11}^* D_{11})^{-1} \{(D_{11}^* C_1 + B_1^* X_\infty)x + D_{11}^* D_{12} u\}.$$

Proof. The condition for part (a) can be shown in the same way as in Chapter 15 and is, in fact, the solution to the FI problem for the general case. We now prove the condition for part (b). It is not hard to see that $\bar{\sigma}(D_{11}) < \gamma$ is necessary since control u cannot feed back w directly, so the D_{11} term cannot be (partially) eliminated as in

the FI problem. The conditions $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty \geq 0)$ can be easily seen as necessary since

$$\sup_{w \in B\mathcal{L}_2[0, \infty)} \min_{u \in \mathcal{L}_2[0, \infty)} \|z\|_2 \leq \min_{u \in \mathcal{L}_2[0, \infty)} \sup_{w \in B\mathcal{L}_2[0, \infty)} \|z\|_2.$$

It is easy to verify directly that the optimal control and the worst disturbance can be chosen in the given form. $\square$

Define

$$\begin{aligned} R_0 &= I - D_{11}^* D_\perp D_\perp^* D_{11} / \gamma^2 \\ \tilde{R}_0 &= I + D_{12}^* D_{11} (\gamma^2 I - D_{11}^* D_{11})^{-1} D_{11}^* D_{12} \\ \hat{R}_0 &= I - D_{11}^* D_{11} / \gamma^2. \end{aligned}$$

Then it is easy to show that

$$\begin{aligned} \|z\|^2 - \gamma^2 \|w\|^2 + \frac{d}{dt}(x^* X x) &= \left\| u + D_{12}^* D_{11} w - \begin{bmatrix} D_{12}^* D_{11} & I \end{bmatrix} F x \right\|^2 \\ &\quad - \gamma^2 \left\| R_0^{1/2} (w - F_{1\infty} x) \right\|^2 \end{aligned}$$

if conditions in (a) are satisfied. On the other hand, we have

$$\|z\|^2 - \gamma^2 \|w\|^2 + \frac{d}{dt}(x^* X x) = \left\| \tilde{R}_0^{1/2} (u - F_{2\infty} x) \right\|^2 - \gamma^2 \left\| \hat{R}_0^{1/2} (w - w_{\text{sfworst}}) \right\|^2$$

if conditions in (b) are satisfied. Integrating both equations from $t = 0$ to ∞ with $x(0) = x(\infty) = 0$ gives

$$\|z\|_2^2 - \gamma^2 \|w\|_2^2 = \left\| u + D_{12}^* D_{11} w - \begin{bmatrix} D_{12}^* D_{11} & I \end{bmatrix} F x \right\|_2^2 - \gamma^2 \left\| R_0^{1/2} (w - F_{1\infty} x) \right\|_2^2$$

if conditions in (a) are satisfied, and

$$\|z\|_2^2 - \gamma^2 \|w\|_2^2 = \left\| \tilde{R}_0^{1/2} (u - F_{2\infty} x) \right\|_2^2 - \gamma^2 \left\| \hat{R}_0^{1/2} (w - w_{\text{sfworst}}) \right\|_2^2$$

if conditions in (b) are satisfied. These relations suggest that an optimal control law and a worst disturbance for problem (a) are

$$u = -D_{12}^* D_{11} w + \begin{bmatrix} D_{12}^* D_{11} & I \end{bmatrix} F x, \quad w = F_{1\infty} x$$

and that an optimal control law and a worst disturbance for problem (b) are

$$u = F_{2\infty} x \quad w = w_{\text{sfworst}}.$$

Moreover, if problem (b) has a solution for a given γ , then the corresponding differential game problem

$$\min_{u \in \mathcal{L}_2[0, \infty)} \sup_{w \in \mathcal{L}_2[0, \infty)} \{ \|z\|_2^2 - \gamma^2 \|w\|_2^2 \}$$

has a saddle point, i.e.,

$$\sup_{w \in \mathcal{L}_2[0, \infty)} \min_{u \in \mathcal{L}_2[0, \infty)} \{ \|z\|_2^2 - \gamma^2 \|w\|_2^2 \} = \min_{u \in \mathcal{L}_2[0, \infty)} \sup_{w \in \mathcal{L}_2[0, \infty)} \{ \|z\|_2^2 - \gamma^2 \|w\|_2^2 \}$$

since the solvability of problem (b) implies the solvability of problem (a). However, the converse may not be true. In fact, it is easy to construct an example so that the problem (a) has a solution for a given γ and problem (b) does not. On the other hand, the problems (a) and (b) are equivalent if $D_{11} = 0$.

17.9 Parameterization of State Feedback $\mathcal{H}_\infty$ Controllers

In this section, we shall consider the parameterization of all state feedback control laws. We shall first assume for simplicity that $D_{11} = 0$ and show later how to reduce the general $D_{11} \neq 0$ case to an equivalent problem with $D_{11} = 0$. We shall assume

$$G = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & 0 & D_{12} \\ I & 0 & 0 \end{array} \right].$$

Note that the state feedback $\mathcal{H}_\infty$ problem is *not* a special case of the output feedback problem since $D_{21} = 0$. Hence the parameterization cannot be obtained from the output feedback.

Theorem 17.7 *Suppose that the assumptions (AS1) – (AS3) are satisfied and that B_1 has the following SVD:*

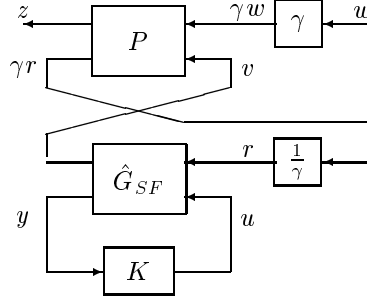
$$B_1 = U \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} V^*, \quad UU^* = I_n, \quad V^*V = I_{m_1}, \quad 0 < \Sigma \in \mathbb{R}^{r \times r}.$$

There exists an admissible controller $K(s)$ for the SF problem such that $\|T_{zw}\|_\infty < \gamma$ if and only if $H_\infty \in \text{dom}(\text{Ric})$ and $X_\infty = \text{Ric}(H_\infty) \geq 0$. Furthermore, if these conditions are satisfied, then all admissible controllers satisfying $\|T_{zw}\|_\infty < \gamma$ can be parameterized as

$$K = F_\infty + \left\{ I_{m_2} + Q \begin{bmatrix} \Sigma^{-1} & 0 \\ 0 & I_{m_1-r} \end{bmatrix} U^{-1} B_2 \right\}^{-1} Q \begin{bmatrix} \Sigma^{-1} & 0 \\ 0 & I_{m_1-r} \end{bmatrix} U^{-1} (sI - \hat{A})$$

where $F_\infty = -(D_{12}^ C_1 + B_2^* X_\infty)$, $\hat{A} = A + \gamma^{-2} B_1 B_1^* X_\infty + B_2 F_\infty$, and $Q = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \in \mathcal{RH}_2$ with $\|Q_1\|_\infty < \gamma$. The dimensions of Q_1 and Q_2 are $m_2 \times r$ and $m_2 \times (m_1 - r)$, respectively.*

Proof. The conditions for the existence of the state feedback control law have been shown in the last section and in Chapter 15. We only need to show that the above parameterization indeed satisfies the $\mathcal{H}_\infty$ norm condition and gives all possible state feedback $\mathcal{H}_\infty$ control laws. As in the proof of the FI problem in Chapter 16, we make the same change of variables to get $\hat{G}_{SF}$ instead of $\hat{G}_{FI}$:



$$\hat{G}_{SF} = \left[\begin{array}{c|cc} A_{tmp} & B_1 & B_2 \\ \hline -F_\infty & 0 & I \\ I & 0 & 0 \end{array} \right].$$

So again from Theorem 16.2 and Lemma 16.8, we conclude that K is an admissible controller for G and $\|T_{zw}\|_\infty < \gamma$ iff K is an admissible controller for $\hat{G}_{SF}$ and $\|T_{vr}\|_\infty < \gamma$. Now let $L = B_2 F_\infty$, and then $A_{tmp} + L = A_{tmp} + B_2 F_\infty$ is stable. All controllers that stabilize $\hat{G}_{SF}$ can be parameterized as $K = \mathcal{F}_\ell(M_{tmp}, \Phi)$, $\Phi \in \mathcal{RH}_\infty$ where

$$M_{tmp} = \left[\begin{array}{c|cc} A_{tmp} + B_2 F_\infty + L & -L & B_2 \\ \hline F_\infty & 0 & I \\ -I & I & 0 \end{array} \right].$$

Then $T_{vr} = \mathcal{F}_\ell(\hat{G}_{SF}, \mathcal{F}_\ell(M_{tmp}, \Phi)) =: \mathcal{F}_\ell(N_{tmp}, \Phi)$. It is easy to show that

$$N_{tmp} = \left[\begin{array}{c|cc} A_{tmp} + B_2 F_\infty & B_1 & 0 \\ \hline -F_\infty & 0 & I \\ I & 0 & 0 \end{array} \right].$$

Now let $\Phi = F_\infty + \hat{\Phi}$, and we have

$$\mathcal{F}_\ell(N_{tmp}, \Phi) = \hat{\Phi} \left[\begin{array}{c|c} A_{tmp} + B_2 F_\infty & B_1 \\ \hline I & 0 \end{array} \right].$$

Let

$$\hat{\Phi} = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \begin{bmatrix} \Sigma^{-1} & 0 \\ 0 & I \end{bmatrix} U^{-1}(sI - \hat{A}).$$

Then the mapping from $Q = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix} \in \mathcal{RH}_2$ to $\hat{\Phi} \in \mathcal{RH}_\infty$ is one-to-one. Hence we have

$$\mathcal{F}_\ell(N_{tmp}, \Phi) = \begin{bmatrix} Q_1 & 0 \end{bmatrix} V^*$$

and $\|\mathcal{F}_\ell(N_{tmp}, \Phi)\|_\infty = \|Q_1\|_\infty$, which in turn implies that $\|T_{vr}\|_\infty = \|\mathcal{F}_\ell(N_{tmp}, \Phi)\|_\infty < \gamma$ if and only if $\|Q_1\|_\infty < \gamma$. Finally, substituting $\Phi = F_\infty + \hat{\Phi}$ into $K = \mathcal{F}_\ell(M_{tmp}, \Phi)$, we get the desired controller parameterization. $\square$

The controller parameterization for the general case can also be obtained:

$$G_g(s) = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ I & 0 & 0 \end{array} \right].$$

Let

$$N = \begin{bmatrix} -D_{11} & (I - D_{11}D_{11}^*)^{1/2} \\ (I - D_{11}^*D_{11})^{1/2} & D_{11}^* \end{bmatrix},$$

and then $N^*N = I$. Furthermore, by Theorem 16.2, we have that K stabilizes G_g and $\|\mathcal{F}_\ell(G_g, K)\|_\infty < 1$ if and only if $\mathcal{F}_\ell(G_g, K)$ internally stabilizes N and

$$\|\mathcal{F}_\ell(N, \mathcal{F}_\ell(G_g, K))\|_\infty < 1.$$

Note that

$$\mathcal{F}_\ell(N, \mathcal{F}_\ell(G_g, K)) = \mathcal{F}_\ell(M, \tilde{K})$$

with

$$M(s) = \left[\begin{array}{c|cc} A + B_1D_{11}^*R_1^{-1}C_1 & B_1(I - D_{11}^*D_{11})^{-1/2} & B_2 + B_1D_{11}^*R_1^{-1}D_{12} \\ \hline R_1^{-1/2}C_1 & 0 & R_1^{-1/2}D_{12} \\ I & 0 & 0 \end{array} \right]$$

where $R_1 := I - D_{11}D_{11}^*$. In summary, we have the following lemma.

Lemma 17.8 *There is a K that internally stabilizes G_g and $\|\mathcal{F}_\ell(G_g, K)\|_\infty < 1$ if and only if $\|D_{11}\| < 1$ and there is a $\tilde{K}$ that stabilizes M and $\|\mathcal{F}_\ell(M, \tilde{K})\|_\infty < 1$.*

Now Theorem 17.7 can be applied to the new system $M(s)$ to obtain the controller parameterization for the general problem with $D_{11} \neq 0$.

17.10 Notes and References

The detailed derivation of the $\mathcal{H}_\infty$ solution for the general case is treated in Glover and Doyle [1988, 1989]. The loop-shifting procedures are given in Safonov, Limebeer, and Chiang [1989]. The idea is also used in Zhou and Khargonekar [1988] for state feedback problems. A fairly complete solution to the singular $\mathcal{H}_\infty$ problem is obtained in Stoorvogel [1992]. The $\mathcal{H}_\infty$ filtering and smoothing problems are considered in detail in Nagpal and Khargonekar [1991]. The Youla parameterization approach is treated very extensively in Doyle [1984] and Francis [1987] and in the references therein. The presentation of the state feedback $\mathcal{H}_\infty$ control in this chapter is based on Zhou [1992].